

$$-0.0455 \quad \varphi_1 - 0.0625 \quad \varphi_2 + 0.1413 \quad \varphi_3 - 0.0333 \quad \varphi_4 = 6.25$$

$$-0.04 \quad \varphi_1 - 0.1 \quad \varphi_2 - 0.0333 \quad \varphi_3 + 0.1733 \quad \varphi_4 = -8.6$$

Где

$$G_2 = \frac{1}{R_2} = \frac{1}{20} = 0.05 \text{ S}$$

$$G_1 = \frac{1}{R_1} = \frac{1}{16} = 0.0625 \text{ S}$$

$$G_5 = \frac{1}{R_5} = \frac{1}{22} = 0.0454545 \text{ S}$$

$$G_6 = \frac{1}{R_6} = \frac{1}{30} = 0.0333333 \text{ S}$$

$$G_3 = \frac{1}{R_3} = \frac{1}{25} = 0.04 \text{ S}$$

$$G_4 = \frac{1}{R_4} = \frac{1}{18} = 0.0555556 \text{ S}$$

Величины системы с помощью матрицы

$$\varphi = (AG_B A^T)^{-1} (AJ_B - AG_B E_B)$$

После расчета получим

$$\varphi_1 = 110 \text{ V}$$

$$\varphi_2 = 105.95 \text{ V}$$

$$\varphi_3 = 141.622 \text{ V}$$

$$\varphi_4 = 64.129 \text{ V}$$

Определим токи ветвей

$$I_2 = \frac{\varphi_2 - \varphi_1 + E_1}{R_2} = \frac{105.95 - (110) + 100}{20} = 4.797 \text{ A}$$

$$I_1 = \frac{\varphi_1 - \varphi_2 + E_1}{R_1} = \frac{105.95 - (141.622) + 100}{16} = 4.02 \text{ A}$$

$$I_0 = (\varphi_2 - \varphi_4) G_0 = (105.95 - (64.129)) \cdot 0.1 = 4.182 \text{ A}$$

$$J_0 = J_0 = 13 \text{ A}$$

$$I_5 = \frac{\varphi_1 - \varphi_3}{R_5} = \frac{110 - (141.622)}{22} = -1.437 \text{ A}$$

$$I_6 = \frac{\varphi_2 - \varphi_4}{R_6} = \frac{105.95 - (64.129)}{30} = 2.583 \text{ A}$$

$$I_3 = \frac{\varphi_1 - \varphi_4 + E_3}{R_3} = \frac{110 - (64.129) + 110}{25} = 6.235 \text{ A}$$

$$I_4 = \frac{\varphi_3 - \varphi_1 + E_4}{R_4} = \frac{0 - (110) + 110}{18} = 0 \text{ A}$$

2 Рассчитайте токи во всех ветвях ветвей контурных токов